

A Numerical Algorithm for simulating multiphase flows: ice, melt, vapor, deformable solids, high speed, and low speed flows *

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ABSTRACT: A new Eulerian method is presented for simulating complex multiphase flows consisting of any number of types of materials, to include compressible fluids, incompressible fluids, ice, melt, vapor, and solids undergoing plastic deformation. The interfaces between materials is modeled and discretized as sharp.

1 Mathematical Modeling

There are M materials and each material $1 \leq m \leq M$ occupies the region,

$$\Omega_m(t) = \{(\vec{x}, t) | \phi_m(\vec{x}, t) \geq 0\} \quad (1)$$

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The $\phi_m(\mathbf{x}, t)$ are level set functions[2, 1]. The zero level set of such functions demarcate the boundary of material m :

$$\partial\Omega_m(t) = \{(\vec{x}, t) | \phi_m(\vec{x}, t) = 0\} \quad (2)$$

5 In the bulk regions of material m we have the following equations for compressible materials:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \quad (3)$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p \mathbb{I} - \tau) = \vec{F}_{body} \quad (4)$$

$$(\rho E)_t + \nabla \cdot ((\rho E + p) \vec{u} - \vec{u} \cdot \tau - k \nabla T) = \vec{u} \cdot \vec{F}_{body} \quad (5)$$

$$(\rho Y)_t + \nabla \cdot (\rho Y \vec{u}) = \nabla \cdot (\rho D \nabla Y) \quad (6)$$

$$\tau = 2\mu \bar{D} \quad (7)$$

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$$\bar{D} = D - \frac{\text{Trace}(D)\mathbb{I}}{\text{Trace}(\mathbb{I})} \quad (8)$$

$$D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \quad (9)$$

$$\rho E = \rho e_{\text{int}} + \frac{1}{2} \rho |\vec{u}|^2 \quad (10)$$

$$p = p(\rho, e) \quad e = c_v T \quad (11)$$

If the material is a viscoelastic FENE-CR material, then one has the additional
15 equation for the configuration tensor A ,

$$A_t + \vec{u} \cdot \nabla A = A \cdot (\nabla \vec{u})^T + \nabla \vec{u} \cdot A - \frac{f(R)}{\lambda} (A - \mathbb{I}), \quad (12)$$

and the momentum constitutive law term is modified,

$$\tau = \tau + \frac{\tilde{\eta}_P}{\lambda} f(R) A. \quad (13)$$

If the material is an elastic material undergoing plastic deformation, then one has the following additional equation for the deviatoric stress tensor,

$$s_t + \vec{u} \cdot \nabla s + s \Omega - \Omega s = 2G(\bar{D} - D^p), \quad (14)$$

and the momentum constitutive law term is modified,

$$\tau = \tau + s. \quad (15)$$

20 If the material is an elastic material undergoing plastic deformation, then the plastic strain equation is solved too,

$$\epsilon_t^p + \vec{u} \cdot \nabla \epsilon^p = \sqrt{\frac{2}{3}} |\Lambda|. \quad (16)$$

$$|\Lambda| = \sqrt{\frac{3}{2}} \sqrt{D^p : D^p} \quad (17)$$

$$D^p = \Lambda N \quad N = \sqrt{3/2} \frac{s}{S_e} \quad (\dot{\epsilon}^p)^2 = \frac{2}{3} \Lambda^2 \quad (18)$$

Ω is the spin tensor,

$$\Omega_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right) \quad (19)$$

25 The effective stress is defined as,

$$S_e = \sqrt{\frac{3}{2}} \sqrt{s : s} \quad (20)$$

The Johnson-Cook model yield stress is given by,

$$\sigma_y = (A + B(\epsilon^p)^n)(1 + C \ln(\frac{\dot{\epsilon}^p}{\dot{\epsilon}_0}))(1 - \theta^m) \quad \theta = \frac{T - T_0}{T_m - T_0} \quad (21)$$

One form of the Zerilli Armstrong model yield stress is given by,

$$\beta = \beta_0 - \beta_1 \log(\dot{\epsilon}^p / \dot{\epsilon}_{ref}^p) \quad (22)$$

$$\sigma_y = A + B(\epsilon^p)^n e^{-\beta T} \quad (23)$$

The energy contribution due to elasto-plastic deformation is modeled as,

$$\beta \dot{\epsilon}^p S_e, \quad (24)$$

30 and is added to (5). In order to determine Λ , as it appears in (18), the following “radial return algorithm” is implemented (see [3] for the case when $n = 1$ in (21) or (23)):

1. $\dot{\epsilon}^p = 0, k = 0$

2. $S_e = \sqrt{\frac{3}{2}} \sqrt{s : s}$

- 35 **3.** $\sigma_y = \sigma_y(T, \epsilon^p, \dot{\epsilon}^p)$
- 4.** if $S_e - \sigma_y \leq 0$ then $\Lambda = 0$ (i.e. $D^p : D^p = 0$).
- 5.** if $S_e > \sigma_y$ then,
- 6.** Solve the following equation for ϵ_{new}^p :

$$S_e - \sigma_y = 3G(\epsilon_{new}^p - \epsilon^p) + B((\epsilon_{new}^p)^n - (\epsilon^p)^n) \quad (25)$$

7.

$$\dot{\epsilon}^p = \frac{\epsilon_{new}^p - \epsilon^p}{\Delta t} \quad (26)$$

8.

$$\Lambda = \sqrt{\frac{3}{2}} \dot{\epsilon}^p \quad (27)$$

$$D^p = \Lambda N \quad (28)$$

40 **9.** $k = k + 1$

10. if $k = 1$ then go back to step 3.

The mathematical model at sharp interfaces is given as follows. First, define a “closest point” function:

$$\vec{x}_c(\vec{x}) = \operatorname{argmin}_{\vec{y}, m} \{ ||\vec{y} - \vec{x}|| | \phi_m(\vec{y}) = 0 \} \quad (29)$$

45 Taking into account mass transfer, a source material’s level set function (e.g. ice that is melting or liquid that is turning into vapor) satisfies

$$(\phi_{m_s})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla \phi_{m_s} = - \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}} |\nabla \phi_{m_s}| \quad (30)$$

The destination material’s level set function satisfies

$$(\phi_{m_d})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla \phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}} |\nabla \phi_{m_d}| \quad (31)$$

The above equations are equivalent to the following:

$$(\phi_{m_s})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla \phi_{m_s} = - \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}} |\nabla \phi_{m_s}| \quad (32)$$

$$(\phi_{m_d})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla \phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}} |\nabla \phi_{m_d}| \quad (33)$$

Now, we define the interface normal:

$$\vec{n}_m = \frac{\nabla \phi_m}{\|\nabla \phi_m\|} \quad (34)$$

50 The interface curvature is,

$$\kappa_m = \nabla \cdot \vec{n}_m \quad (35)$$

The rate of mass transfer is,

$$\dot{m} = \frac{k_{m_s} \nabla T_{m_s} - k_{m_d} \nabla T_{m_d}}{L} \cdot \vec{n}_{m_d}. \quad (36)$$

The phase change interface temperature condition is,

$$T_{m_s} = T_{m_d} = T_{\text{saturation}}(P_{\text{equilibrium}}) \quad (37)$$

For evaporation, the following rate of mass transfer should match that in (36):

$$\dot{m} = \frac{-\vec{n}_{m_s} \cdot \nabla Y_{m_d} \rho_{m_d} D}{Y^\Gamma - 1} \quad (38)$$

55 Let the tensor σ represent pressure, viscosity, viscoelastic, and elastic contributions to the momentum equations. One has:

$$\sigma = -p\mathbb{I} + \tau. \quad (39)$$

The stress jump conditions are

$$\vec{n}_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = -\gamma_{m_s, m_d} \kappa_{m_s} - \dot{m}^2 \left(\frac{1}{\rho_{m_s}} - \frac{1}{\rho_{m_d}} \right) \quad (40)$$

$$(\vec{t}_{i, \text{tangent}})_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = 0 \quad i = 1, 2 \quad (41)$$

The velocity jump conditions are

$$(\vec{u}_{m_s} - \vec{u}_{m_d}) \cdot \vec{n}_{m_s} = \dot{m} \left(\frac{1}{\rho_{m_d}} - \frac{1}{\rho_{m_s}} \right) \quad (42)$$

2 phase change algorithm

60 See the following articles and below: [5], [4], [6].

1. For each cell in a narrow band about the interface, use the closest point map to find the corresponding closest point on the interface. Approximate $\dot{m}$ at the closest point.
2. extend $\dot{m}$ further into bulk regions.

- 65 **3.** If distributing $\dot{m}$ to the destination material side, then the velocity on the interface corresponds to the source material velocity and the interface speed is:

$$\vec{u}_{source} + \frac{|\dot{m}|}{\rho_s} \vec{n}_s \quad (43)$$

- If distributing $\dot{m}$ to the source material side, then the velocity on the interface corresponds to the destination material velocity and the interface speed is:
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$$\vec{u}_{dest} + \frac{|\dot{m}|}{\rho_d} \vec{n}_s \quad (44)$$

4. Using step 3 as a guide, do unsplit CISL phase change advection.
 5. The difference in volume fraction from step 4 gives the $\dot{m}$ expansion source term to be applied during the pressure update step.
 6. The $\dot{m}$ terms are distributed to one side of the interface. If distributed to the destination material side, then distribute where $F_{dest} > 1/2$. This guarantees that all faces with at least one adjoining source material cell will have the source material velocity defined at that face. Please refer to the diagram.
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For compressible flow, with a predictor and a corrector step, there is a problem. Suppose evaporation with $\dot{m}$ distributed to the vapor (destination). Please refer to the diagram. Also, in words:

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1. During the predictor step, there is no problem because the advection velocity at time t^n is used and all t^n source (liquid) material faces have the source material velocity.
 2. During the corrector step, there is a problem because the advection velocity at time t^{n+1} is used which has $\dot{m}$ distributed to the vapor side of the t^{n+1} interface. But cells on the vapor side at t^{n+1} might have been liquid cells at t^n ! Now, the discretization is inconsistent because the interface is not wholly advected by the source (liquid) velocity.
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90 **3 Applications**

1. Jet in cross flow problem (military jet engines, Pratt and Whitney)
2. Diesel injectors
3. Rotating Detonation Engines (see latest article with Everett Wenzel and Marco Arienti).

95 4 Algorithm

1. Cell integrated semi Lagrangian advection
2. Cell integrated semi Lagrangian advection due to phase change.
3. Viscosity, Viscoelastic, and elastic forces.
4. Temperature Diffusion
- 100 5. Mass Fraction Diffusion
6. Pressure Gradient
7. Go back to step 1.

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